

ITEC 3100 – Introduction to Networks

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Lab 7:

Please use a different color font for your answers so I can clearly see them:

Watch this OSI Model video:

<https://www.professormesser.com/network-plus/n10-007/understanding-the-osi-model-2/>

Q1. What is ISO and how many countries does it represent?

International Organization for Standardization

More than 160 countries

Q2. What is the purpose of the OSI model?

Helps for troubleshooting, what things are

Q3. List the 7 layers of the OSI model both name and number from top to bottom:

7- Application

6-Presentation

5-Session

4-Transport

3-Network

2-Data link

1-Physical

Q4. Which layer is responsible for converting the data or formatting? **Presentation**

Q5. Which layer is responsible for breaking the data into smaller “segments”?

Transport

Q6. Which layer deals with “packets”?

Network

Q7. Which layer does your browser operate at?

Application

Q8. Which layer deals with the creating on-going communications?

Session

Q9. Which layer would you find IP addresses? Which layer would you find MAC addresses?

IP-Networks

Mac- Data link

Q10. What is a protocol? When you run the ping command which protocol are you using?

Allows computers to communicate over the network

ICMP

Q11. Which layers do the following hardware devices operate at?

- a) Hub-Physical
- b) Router-Network
- c) Switch-Data link
- d) Repeater-Physical
- e) Gateway-Data link
- f) Computers-Application

Q12. Which layer do the following protocols operate at?

- a) DHCP- application
- b) FTP- application
- c) TCP- transport
- d) DNS- application
- e) UDP- transport
- f) HTTPS- application
- g) IP- network
- h) ICMP- network
- i) HTTP- application

Q13. Is the OSI model a theoretical or practical concept?

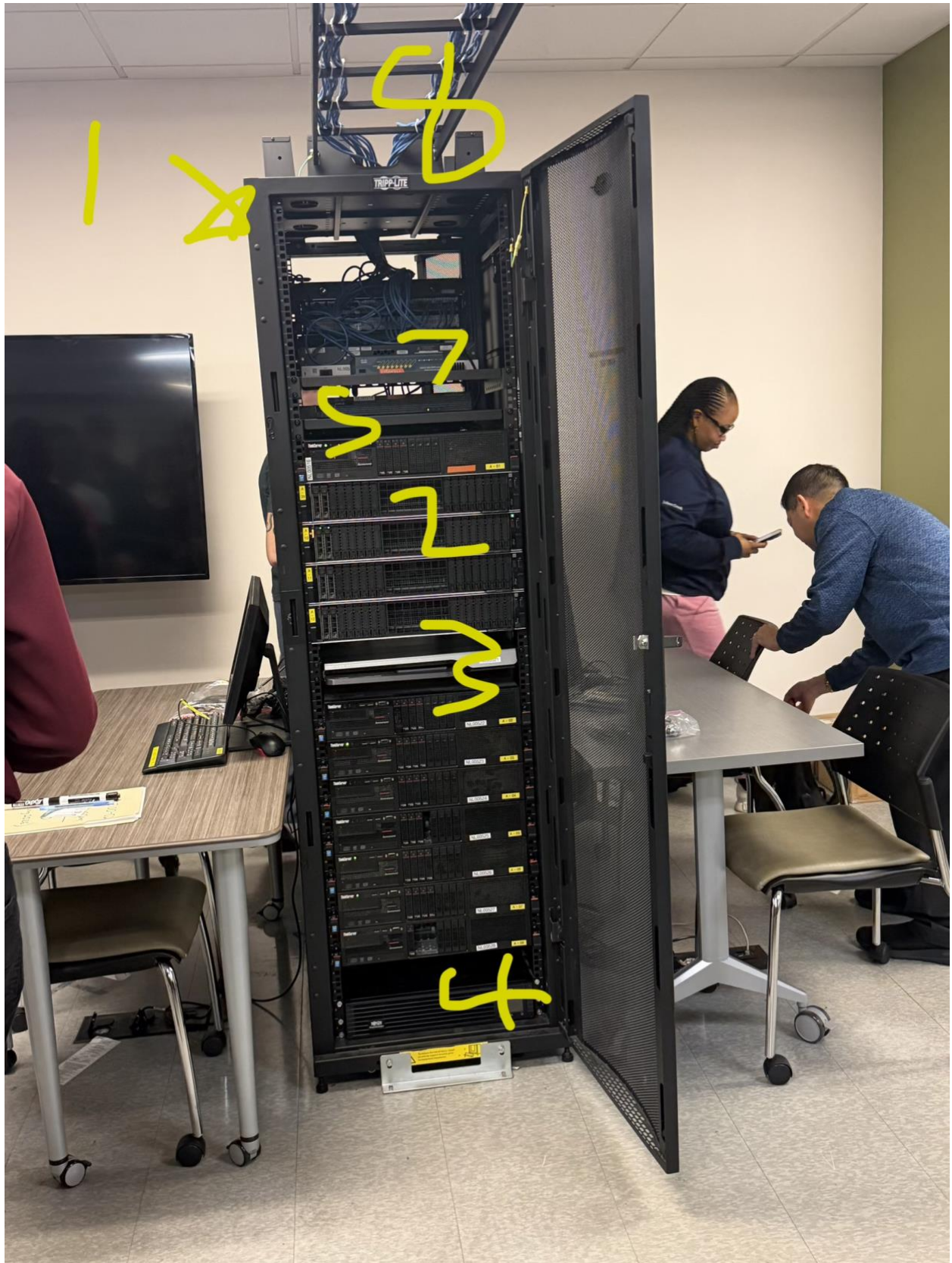
Theoretical

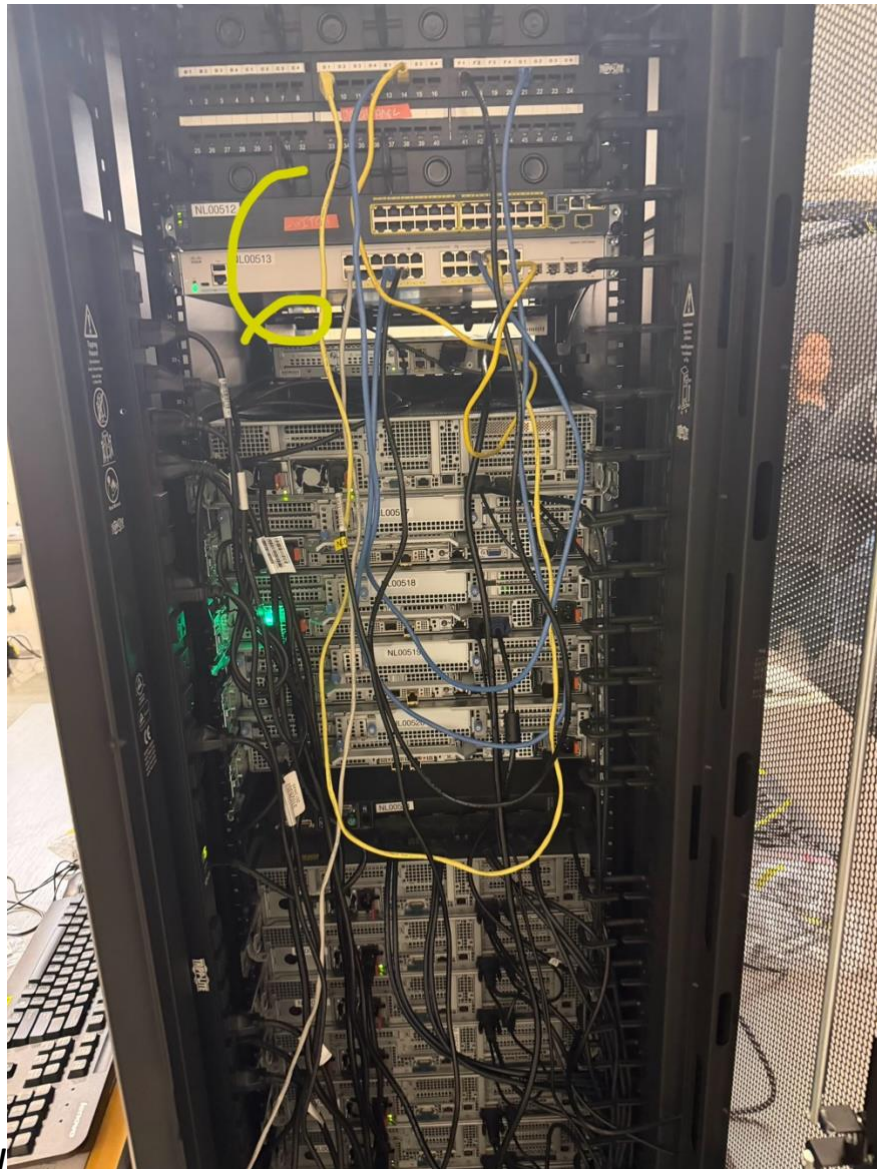
Q14. What is the process called when more headers are added to the data units as it goes down each layer?

encapsulation

Q15. Insert pictures taken from the Network Lab and label for the following:

- 1) Server Rack
- 2) Servers
- 3) KVM
- 4) UPS
- 5) Router
- 6) Switch
- 7) Firewall
- 8) Cable connections





Complete and upload to the Lab 7 dropbox.